Independence:

→ A, B are independent "events"

$$P(A \cap B) = P(A)\,P(B)$$

A = { HHH, TTT } ← all tosses have same result

B = { HTT, HTH, HHT, THH, TTH, THT } ← atleast one H & atleast one T

C = { TTT, HTT, THT, TTH } ← atmost one H

$\Omega = \{H,T\}^3$, $|\Omega| = 2^3 = 8$

(a) A, B are independent ?

$$P(A \cap B) \stackrel{?}{=} P(A)\,P(B)$$

$$0 \stackrel{?}{=} \frac{2}{8} \cdot \frac{6}{8} \quad \times$$

A, B are not independent

(b) B, C are independet ?

$P(B \cap C) = \frac{3}{8}$ ← same as this

$P(B) = \frac{6}{8} = \frac{3}{4}$

$P(C) = \frac{4}{8} = \frac{1}{2}$

$P(B)\,P(C) = \frac{3}{8}$

B, C independet

(C) A, C are independet ?

"can be done similarly"

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Q4] A, B are independet (given)

show, A^c and B^c are independt

$P(A \cap B) = P(A)\,P(B)$ ✓(given)

$P(A^c \cap B^c) \stackrel{?}{=} P(A^c)\,P(B^c)$

$A^c \cap B^c = (A \cup B)^c$ "De Morgan's law"

$P((A \cup B)^c) = 1 - P(A \cup B)$

$= 1 - (P(A) + P(B) - P(A \cap B))$

$= 1 - P(A) - P(B) + P(A)\,P(B)$

$= (1 - P(A))(1 - P(B))$

$P(A^c \cap B^c) = P(A^c)\,P(B^c)$, Thus independent

13] N coin.. $P(H) = p$

$A_k \rightarrow$ k^{th} toss: head, $(k-1)^{th}$ toss: tail $\rightarrow$ reward in k^{th} step

(a) A_k, A_{k+1} independent ?

$\{ _ _ _ \}$ (k-1, k, k+1)

$A_k = \{ THH, THT \}$

$A_{k+1} = \{ TTH, HTH \}$

$\star\ P(A_k \cap A_{k+1}) \stackrel{?}{=} P(A_k)\,P(A_{k+1})$ ✓ ✓

$A_k \cap A_{k+1} = \{\} = \emptyset$

$P(A_k \cap A_{k+1}) = 0 \neq x > 0$

NOT INDEPENDENT

Independence (other definitions)

2. $P(A \mid B) = P(A)$

3. $P(B \mid A) = P(B)$

	k-1	k	k+1
A_k	T	H	*
A_{k+1}	*	T !	H ✓

$(1-p)\ p$

$(1-p)p$

A_{k+1}, A_k are independet ?

Is $P(A_{k+1} / A_k) \stackrel{?}{=} P(A_{k+1})$

$0 \stackrel{?}{=} (1-p)p$

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NOT INDEPENDENT

(b) A_k , A_{k+2} are independent ?

$P(H) = p$
$P(T) = 1-p$

k-1	k	k+1	k+2
T	H	T	H

A_k $\quad$ A_{k+2}

$$P(A_k \cap A_{k+2}) \overset{?}{=} P(A_k)\, P(A_{k+2})$$

$P(A_k \cap A_{k+2})$ = $(1-p)\,p\,(1-p)\,p$ = $p^2(1-p)^2$

$P(A_k) = (1-p)\,p$, $P(A_{k+2}) = (1-p)\,p$ $\Rightarrow$ $p^2(1-p)^2$

A_k, A_{k+2} are independent

Q2] { Alice, Bob, ... others ... } $\longrightarrow$ 8 people.

* '3' are randomly chosen

A $\rightarrow$ alice is one of them

B $\rightarrow$ Bob "

* $P(A)$, $P(A|B)$, $P(A \cup B)$

(i) $P(A)$:

8 people $\rightarrow$ 3 chosen

$\Omega = \{ \ldots \}$, $|\Omega| = {}^8C_3$

$A = \{ \ldots \}$, $|A| = {}^7C_2$

$$P(A) = \frac{{}^7C_2}{{}^8C_3}$$

(ii) $P(A|B) = P(A \cap B) / P(B)$

- $P(A \cap B) = \dfrac{{}^6C_1}{{}^8C_3}$

$\underline{A}, \underline{B}, _$ ↙ 6 of 8

- $P(B) = \dfrac{{}^7C_2}{{}^8C_3}$

$\Rightarrow P(A|B) = {}^6C_1 \,/\, {}^7C_2$ //

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(2)

★ $P(A|B)$

↘ $P(\cdot|B)$: B, _, _ $\Rightarrow$ ${}^7C_2$ total —,— can be filled

$P(A|B)$: B, A, _ $\Rightarrow$ ${}^6C_1$ ways

Ans: ${}^6C_1 / {}^7C_2$

(3)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

— "can be done similarly"

★ ${}^7C_2$, ${}^mC_n = \frac{m!}{(m-n)!\,n!}$

$= \frac{7!}{5!\,2!} = \frac{5 \cdot 6 \cdot 7}{5!\,2!} = \frac{6 \cdot 7}{2!} = 21$

${}^{10}C_3 = \frac{8 \cdot 9 \cdot 10}{3!}$ Simpler way to calculate ${}^mC_n$

Q6]

- 4 instructors : 1, 2, 3, 4
- $P(\text{any one}) = 1/4$
- N classes

$P(\text{all instructors taught atleast once})$

$A_1^c = \{1 \text{ never taught}\}$

$A_1 = \{1 \text{ taught atleast once}\}$

$A_2 = \{2 \quad " \quad\}$

$A_3 = \{3 \quad " \quad\}$

$A_4 = \{4 \quad " \quad\}$

$i = 1$

Total no. of ways: 4^N

→ # 1 never teaches : 3^N

3, 4 never teaches : 2^N

We want,

$$P(A_1 \cap A_2 \cap A_3 \cap A_4) = 1 - P(A_1^c \cup A_2^c \cup A_3^c \cup A_4^c)$$

- $P(A_1^c \cup A_2^c \cup A_3^c \cup A_4^c)$

$$= \sum_{i=1}^{4} P(A_i^c) - \sum_{i \neq j} P(A_i^c \cap A_j^c) + \cdots$$

i and j

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A_i^c = { i never taught }

$= \sum_{i=1}^{4} \left(\frac{3}{4}\right)^N$

3/4 3/4 3/4

1 / 2 / 3 / ... / N

'i' never taught

$- \sum_{i \neq j} \left(\frac{2}{4}\right)^N + \sum_{i \neq j \neq k \neq i} \left(\frac{1}{4}\right)^N$

$= 4\left(\frac{3}{4}\right)^N - {}^4C_2\left(\frac{2}{4}\right)^N + {}^4C_3\left(\frac{1}{4}\right)^N$ //

What if 'K' instructions? (K=4) above

$\Rightarrow {}^KC_1\left(\frac{K-1}{K}\right)^N - {}^KC_2\left(\frac{K-2}{K}\right)^N + \cdots$

$P(A_1^c \cap A_2^c \cap \cdots A_k^c)$ Ans: $1 - P(\cdots)$

Q14]

A	B	C
↓	↓	↓
P(H) = 0.5	0.7	0.9

5 coins

2A 2B 1C

$\star\ P(B/A) = \frac{P(A/B)\,P(B)}{P(A)}$

$\star\ P(A/H)$

$= \frac{P(H/A)\,P(A)}{P(H)}$

- $P(A) = \frac{2}{5}$, $P(H/A) = 0.5$,
- $P(H) = ?$

$= P(H/A)P(A) + P(H/B)P(B) + P(H/C)P(C)$

$= (0.5)\left(\frac{2}{5}\right) + (0.7)\left(\frac{2}{5}\right) + (0.9)\left(\frac{1}{5}\right)$

$P(X) = P(X|A_1)\,P(A_1) + P(X|A_2)\,P(A_2) \cdots$

given $A_1 \cup A_2 \cup A_3 \ldots A_N = \Omega$,

$A_1, A_2 \ldots A_N$ are disjoint, $A_i \cap A_j = \phi,\ i \neq j$

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Q24)

$$P(A) = \frac{1}{4}, \quad P(B) = \frac{1}{2}$$

Find maximum & minimum values of $P(A \cup B)$, $P(A \cap B)$

(ii) $P(A \cap B)$

(max) $A \subseteq B$

(i) min : $A \cap B = \emptyset$, $P(A \cap B) \geq 0$

(ii) max : $P(A \cap B) \leq P(A) = \frac{1}{4}$

Therefore $0 \leq P(A \cap B) \leq \frac{1}{4}$

(A) (B) disjoint (min)

(i) $P(A \cup B)$ $\quad 1/4 + 1/2 \quad 0$

$= P(A) + P(B) - P(A \cap B)$

max : $1/4 + 1/2 - 0 = 3/4$

min : $1/4 + 1/2 - 1/4 = 1/2$

Therefore, $\frac{1}{2} \leq P(A \cup B) \leq \frac{3}{4}$

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